MILITARY SPECIFICATION

SOCKS, EXTREME COLD WEATHER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of felt socks furnished in extra-small, small, medium, and large sizes, as specified (see 6.2).

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solitation, form a part of this specification to extent specified herein.

SPECIFICATIONS

FEDERAL

R-N-91	Naphthalene, Technical.
V-T-276	Thread, Cotton.
PPP-B-636	Box, Shipping, Fiberboard.
PPP-T-60	Tape, Packaging, Waterproof.

MILITARY

MIL-T-43566	Tape, Textile, Cotton, General Purpose, Natural or in Colors.

STANDARDS

FEDERAL

FED-STD-191	Textile Test Methods.
FED-STD-751	Stitches, Seams, and Stitchings.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, TX 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8440

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MILITARY

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 Marking for Shipment and Storage
* MIL-STD-147 Palletized Unit Loads

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified.

DRAWING

AIR FORCE

62H4293 Patterns-Socks, Felt, Sizes Extra Small, Small, Medium and Large

* (Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

* 2.1.3 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 Standard Practice for Commercial Packaging.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

2.1.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the test of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.6).

3.1.1 Materials.

3.1.1.1 Felt. Unless otherwise specified the felt for the socks shall be made of 100 percent fleece, 100 percent pulled sheep's wool or a mixture of fleece and pulled sheep's wool. The fleece or wool shall not be lower in grade than the 40's of U.S. standard. The wool shall be of suitable staple length to comply with this specification. Reused wool for garnetted hard ends, sweater clips, and any other similar waste shall not be used.

3.1.1.2 Color. Unless otherwise specified, the color of the felt shall be the natural color of white scoured wool.

3.1.1.3 Physical requirements. The physical requirements of the finished felt shall be as follows.

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Weight per square foot	4 ounces, plus or minus 5 percent
Texture, minimum	
Warp	35 threads per inch
Filling	35 threads per inch
Breaking strength, grab method, minimum	
Warp	165 pounds
Filling	165 pounds
Thickness per square inch with 10-ounce load, minimum	0.170 inch

3.1.2 Tape. The tape shall be 5/8 inch wide and shall conform to type I, class 2 (bleached) of MIL-T-43566.

3.1.3 Thread. The thread used for sewing shall conform to types IA1 and IB1, ticket No. 16, 4 ply of V-T-276.

3.2 Design. The socks shall consist of two quarters and a bottom stitched together. They shall have a tape stay at the front and a tape pull strap at the back.

3.3 Construction. The socks shall be constructed in accordance with Table I, but the manufacturer is not required to follow the exact sequence of operations as listed therein.

TABLE I. Construction.

No.	Description of Operation	Seam Type	Stitch Type
1	CUTTING. Cut the sections of the socks. The edges of all cut sections shall be smooth and in accordance with the dimensions of the patterns furnished by the Government.		
2	ASSEMBLY. Join the quarters at the front. Join the quarters to the bottom section. Join the quarters at the back.	FSa-1	304
3	FRONT STAY. Fold a 1 3/4 inch length of tape across the top of the front seam and box stitch with a box not less than 1/2 inch by 3/8 inch.	FSa-1	301
4	PULL STRAP. Fold in half a 7 1/2 inch length of tape. Place one end of the tape on the inside and the other end on the outside of the back seam and box stitch with a box not less than 3/4 inch by 3/8 inch, making a loop with a 2 1/2 inch opening.	FSa-1	301

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3.3.1 Stitches, seams, and stitchings. Stitches, seams, and stitchings specified herein shall conform to FED-STD-751. The stitches shall be 12 to 14 stitches per inch. Thread conforming to type IA1 of V-T-276 shall be used for the bobbin, and thread conforming to type IB1 of V-T-276 shall be used for the needle for all stitch types.

3.3.1.1 Thread breaks and ends of seams and stitchings. Thread breaks in stitching shall be backstitched not less than 1/2 inch. Ends of seams and stitchings, where not caught in other seams or stitchings, shall be backstitched not less than 1/2 inch. The seams shall be properly sewn so that there will not be any run-offs, twists, pleats, puckers, or open seams.

3.4 Finished measurements and weights. The finished measurements of the socks shall conform to Figure 2. The minimum weight per six pairs of socks shall be 4-1/4 pounds for extra small sizes, 4-1/2 pounds for small sizes, 4-3/4 pounds for medium sizes, 5 pounds for large sizes. A tolerance of plus or minus 1/2 inch for finished measurements is permissible.

3.5 Patterns. Patterns shall be in accordance with drawing 62H4293. Standard patterns will be furnished by the procuring activity and shall be used as a guide in cutting the working patterns. The standard patterns shall not be altered in any manner, and the working patterns shall not be modified without prior approval of the procuring activity.

3.6 Finish. The finished fabric shall be soft and flexible and shall be thoroughly scoured. The face and the back shall be given a good cover nap. Finishing or loading materials shall not be used to increase the weight or the breaking strength of the fabric. The felt shall be free of holes, lumps, or any other defect that might affect its serviceability.

3.7 Pairing. The socks shall be paired and twine passed through the pull straps shall be used to hold the pair of socks together.

3.8 Identification of product. The letters XS, S, M, or L, indicating size, shall be legibly stamped on the pull strap of each sock. Black indelible ink or dry transfer shall be used for the stamping, and the letters shall be 3/8 inch in height and width. In addition, the wool content of the sock shall be stamped adjacent to the size marking. The wool content marking shall be in accordance with the wool products labeling act of 1939.

3.9 Workmanship. Socks shall be clean, well-made, and free of any defects that might affect their appearance or serviceability.

3.10 Reclaimed materiels. The use of reclaimed materiels shall be encouraged to the maximum extent possible.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own

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or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.2.1 Testing of components. In addition to the quality assurance provisions of the subsidiary specifications and drawings, the tests specified in Table II shall be performed on the felt material for the characteristics indicated. The sample unit of felt material shall be 1 yard, full width. The inspection level for all characteristics shall be 4.0 defects per 100 units and shall be applicable to sample unit requirements. A certificate of compliance with the requirements for wool content (see 3.1.1) and thickness of the felt (see 3.1.1.2) will be acceptable.

TABLE II. Testing of Components.

Characteristic (see 3.1.1.2)	Test Method of FED-STD-191
Weight	4041
Texture	
Warp	5050
Filling	5050
Breaking strength	
Warp	5100
Filling	5100

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.4.1 Examination of end item. The finished socks shall be examined for defects. The defects found during this examination shall be classified in accordance with 4.4.2, 4.4.2.1 and 4.4.2.2. The inspection levels shall be II of MIL-STD-105 for 4.4.2 and S-3 of MIL-STD-105 for 4.4.2.1. The acceptable quality level for 4.4.2, shall be 2.5 defects per 100 units for major defects and 10.0 defects per 100 units for total defects. The acceptable quality level for 4.4.2.1 and 4.4.2.2 shall be 4.0 defects per 100 units for total defects. The sample unit shall be one completely fabricated sock, and selection shall be in pairs. The lot size shall be expressed in units of one sock. The sample unit for defects in pairing shall be one pair. Defects of pairing shall be classified as a single defect.

4.4.2 Visual examination. The socks shall be visually examined for defects in finish, design, material, construction, workmanship, and marking. Defects found during this examination shall be classified as follows:

Examine	Defect	Major	Minor
Finish and color (applicable to all components)	Not specified color or definite variation in color		X
	Not soft or flexible	X	
Pairing	Not properly mated, e.g., not right and left on same size	X	
	Badly shaded or wide variation in color		X
	Not secured with twine through the pull straps		X
Cleanliness	Any spot or stain seriously affecting appearance		X
Design	Not in accordance with specified requirements (unless otherwise indicated herein)		X
Material (felt)	Hole, cut, or tear seriously affecting serviceability	X	
	Hole, cut, or tear affecting serviceability but not seriously		X
	Smash, float, slub, abrasion, or any other defect affecting serviceability seriously		X
	Smash, float, slub, abrasion, or any other defect affecting serviceability but not seriously		X
	Not napped on either side	X	
Tape	Not white		X
	Missing	X	

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Examine	Defect	Major	Minor
	Not completely covering seam as required	X	
	Not securely attached, affecting serviceability seriously	X	
	Not securely attached but not affecting serviceability seriously		X
	Not completely box stitched as specified		X
	Lap-over on front stay less than 5/8 inch on either side		X
	Loop on pull strap less than 2 inches or more than 3 inches		X
Construction	Edge of seam overlapping or not tightly butted	X	
	Joining of front or back seams at the bottom section not close and firm (toe or heel), affecting serviceability seriously	X	
	Joining of front or back seams at the bottom section not close and firm (toe and heel) affecting serviceability but not seriously		X
Workmanship (seams and stitching)	Any open seam except front stay or back loop (A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more continuous skipped stitches or run-offs occur.)	X	
	Open seam on front stay or back loop		X
	Twist, pleat, or pucker seriously affecting service-ability	X	
	Twist, pleat, or pucker seriously affecting service-ability, but not seriously		X
	Incorrect stitch or seam type	X	
	Loose stitch tension resulting in a loosely secured seam	X	
	One or two stitches per inch less than specified		X

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Examine	Defect	Major	Minor
	Three or more stitches per inch less than specified	X	
	More than specified stitches per inch, not damaging material		X
	More than specified stitches per inch, damaging material	X	
	Backstitching omitted where required	X	
	Backstitched less than 1/2 inch		X
Marking	Missing, incomplete, incorrect, not legible, not specified size, not in proper location, or not accomplished in the specified manner.		X

4.4.2.1 Examination of finished measurements. The finished socks shall be examined for defects in finished measurements. Any finished measurement not as specified shall be classified as a defect.

4.4.2.2 Examination for weight. The sample used for 4.4.2 shall be used to select the number of bundles designated below. Bundles of socks shall be weighed to comply with weight requirements specified in 3.4. A bundle shall consist of six(6) pairs of wocks. The lot shall be unacceptable if one or more bundles fail to meet the weight requirements.

Sample size (from 4.4.2)	No. of bundles to be weighed
125 or less	3
200 each	4
315 each	5
500 each	6
800 each	7

4.4.3 Examination of preparation for devliery. An examination shall be made to determine if the preservation, packaging, packing, and marking requirements of this specification have been met. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 of MIL-STD-105, and the acceptable quality level shall be 4.0 defects per 100 units. Shipping containers fully prepared for delivery shall be examined for the following defects of closure:

Marking	Omitted; incorrect; illegible; or of improper size, location, sequence, or method of application
Materials	Any component missing
	Any component damaged, affecting serviceability

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Workmanship	Inadequate application of components, such as incomplete closure of case liners, container flaps, loose strapping, or inadequate stapling or bulging or distortion of containers.
Weight	Gross or net weight exceeds requirement.
Content	Number of end items is more or less than required or size on end item does not correspond to size marking of shipping container.
Preservation (Naphthalene Flakes)	Improperly applied or missing.

4.4.4 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level A. Unless previously treated with an acceptable moth-proofing material to protect against moth attack, the socks shall be packaged with naphthalene flakes conforming to class B of R-N-91 scattered around in the unit package. A minimum of 2 ounces of naphthalene flakes shall be used for 1 cubic feet of package volume. Six pairs of socks of one size only shall be neatly stacked in a bundle, with every other pair reversed end for end, and securely tied at each end with a suitable cotton tape or twine.

* 5.1.2 Commercial. Socks shall be packed in accordance with ASTM D 3951.

5.2 Packing.

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5.2.1 Level A. Six bundles (36 pairs) of socks of one size only, packaged as specified in 5.1.1, shall be packed in a box conforming to type CF or SF, class weather-resistant, grade V3c or V3s, style RSC of PPP-B-636. Each shipping container shall be closed and reinforced with flat, steel strapping or tape banding in accordance with the appendix of the applicable container specification. In addition, the corners, the seams, and the manufacturer's joints of the fiberboard boxes shall be sealed with cloth tape. The tape shall be a minimum of 3 inches wide, shall be olive-drab color, and shall conform to class 1 of PPP-T-60.

5.2.2 Level B. Six bundles (36 pairs) of socks of one size only, packaged as specified in 5.1.1, shall be packed in a box conforming to type CF or SF, class domestic, style RSC of PPP-B-636. The minimum bursting strength of the fiberboard shall be 275 pounds per square inch. All flaps of the fiberboard shipping container shall be securely sealed by applying an adhesive, commercially used for the specific product being packed, throughout the entire area of contact between the flaps or by a combination of metal stitching the bottom flaps and sealing the top flaps with adhesive.

* 5.2.3 Commercial. Socks packaged as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, or ASTM D 3951, as applicable.

5.4 Palletization. Socks, packed as specified (see 5.2), shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with K and L or film bonding means O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet patterns used shall be approved by the contracting officer.

6. NOTES

6.1 Intended use. The socks covered by this specification are intended to be worn by personnel of the Air Force and Navy, with the mukluk boot specified in MIL-B-6362.

* 6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.1).

c. When a first article is required (see 3., 4., and 6.).

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* d. Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

e. When palletization is required (see 5.4).

6.3 Figure. Figure 1 is furnished only for information and guidance.

* 6.4 First article. First article samples shall inspected and approved under the appropriate provisions of FAR 52.209. The contracting officer should include specific instructions in all acquisition documents, regarding arrangements for inspection and approval of the first article samples.

6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 99
Navy - NU

Review Activities:
Air Force - 11, [redacted]
DLA - DP

Preparing activity:
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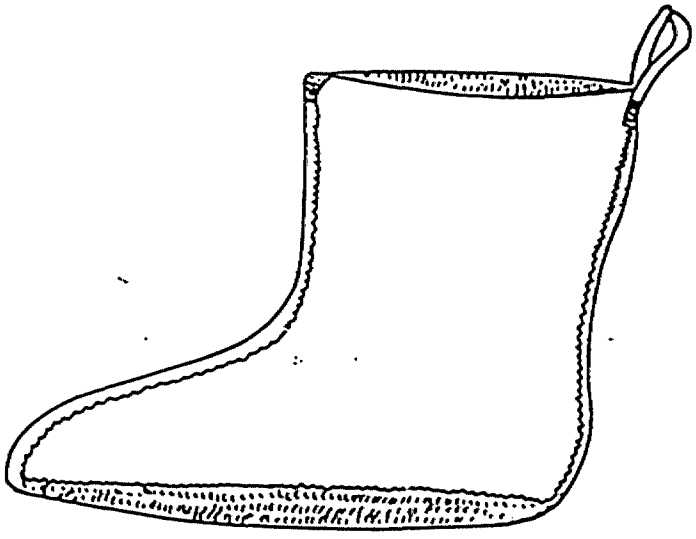
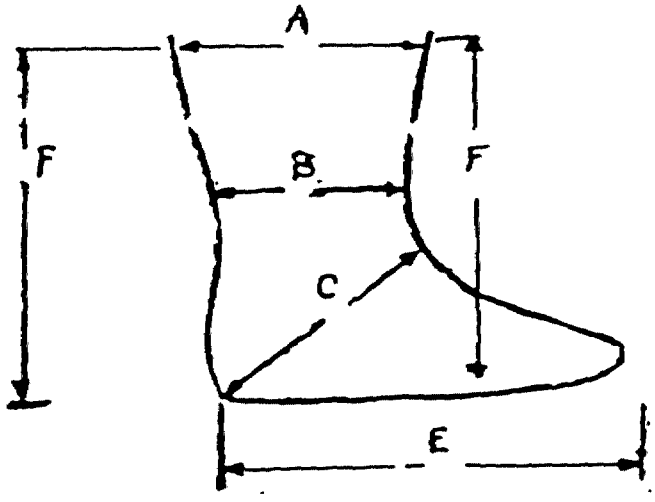
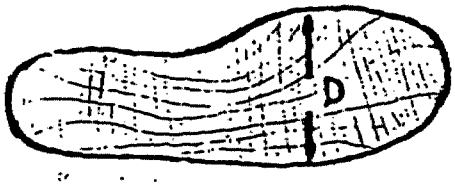


FIGURE 1. Socks, extreme cold weather.

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	Measurement and location	Size			
		Extra small	Small	Medium	Large
		5 to 7	7 1/2 1/	8 to 10 1/	10 1/2 and over 1/
A	Girth at top	15 1/4	15 1/2	16	16 1/2
B	Girth at ankle (4 1/2 inches below top)	13 1/2	14	14	14 1/2
C	Girth, heel to top of instep	16 1/2	16 1/2	17	17 1/2
D	Girth at ball (widest point)	12	12 1/2	13	13 1/2
E	Length, heel to toe	11 1/4	11 1/2	12 1/2	13 1/2
F	Height, back and front	9 1/2	9 1/2	9 1/2	9 1/2

1/ Corresponding shoe size

FIGURE 2. Sock, finished measurements (inches)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-38004C	Socks, Extreme Cold Weather

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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